

Lesson 06 Practice: Answers and Rationales

JKSSB Computer Awareness

October 10, 2026

Provenance: All 20 items are *[Original]* JKSSB-pattern questions for this lesson. No item is a real PYQ, and the answer key is the author's verification of the lesson's concepts.

- Q1. C.** A CPU has few powerful general-purpose cores optimised for latency. A describes a GPU; B and D misdescribe the CPU.
- Q2. B.** A GPU targets parallel throughput work such as graphics and pixels. A names the CPU's general-purpose sequential role; C and D are unrelated to GPU design.
- Q3. B.** Serial processing performs one task at a time, in sequence. A and C describe parallel behaviour; D is false.
- Q4. C.** Parallel processing runs several tasks or operations at the same time. A is the serial definition; B is too narrow; D is false, since parallel work does not always require a GPU.
- Q5. D.** GPU cores are simpler and more numerous, optimised for parallel throughput. A wrongly equates them with CPU cores; B and C misstate their role.
- Q6. B.** GPU stands for Graphics Processing Unit. A, C, and D are invented expansions.
- Q7. A.** CPU stands for Central Processing Unit. B, C, and D are invented expansions.
- Q8. B.** A microprocessor is a CPU implemented on a single chip, and it is general-purpose. A and D name other components; C actually describes a microcontroller.
- Q9. D.** A microcontroller is a microprocessor plus integrated memory and I/O peripherals on one chip. A describes a plain CPU; B and C are unrelated devices.
- Q10. A.** Microcontrollers are used in embedded and special-purpose systems such as appliances and controllers. B, C, and D name other classes of machine.
- Q11. B.** Hyper-threading lets one physical core present two logical threads. It does not double the physical core count (A) or the clock speed (C), and it does not disable multitasking (D).
- Q12. B.** Statements I and II are correct standard descriptions; III is false because GPU cores are simpler and parallel-oriented, not fully general-purpose replacements for CPU cores. A omits II; C includes false III and omits I; D also includes false III.
- Q13. B.** L1 is the smallest and fastest level, closest to the core. A and D confuse the levels or their role; C wrongly places cache on the hard disk.
- Q14. C.** The order from smallest and fastest to largest and slowest is L1, L2, L3. A, B, and D scramble this order.

- Q15. C.** Clock speed in GHz expresses cycles per second. A, B, and D describe storage, memory, or core count, not clock speed.
- Q16. A.** A multi-core processor has multiple independent processing units (cores) on one chip. B, C, and D describe unrelated properties.
- Q17. D.** A single-threaded program uses only one core, and the remaining slowness usually comes from a memory or disk-I/O bottleneck, so the CPU is not automatically at fault. A overstates the benefit of cores; B and C are unrelated to the symptom.
- Q18. D.** A microcontroller is not just a CPU: it integrates memory and I/O on the same chip. A, B, and C are correct statements, so D is the one that is NOT correct.
- Q19. C.** The key difference is the level of integration: the microcontroller includes memory and I/O on the same chip. A and B name secondary attributes; D is false.
- Q20. D.** A GPU suits parallel work such as processing millions of pixels or parallel floating-point operations. A is a sequential general-purpose task (CPU); B and C are not compute-parallel workloads.
- Q21. A.** The 4004 was the first commercial microprocessor. B, C, and D name unrelated hardware or software.
- Q22. A.** GPUs handle many parallel threads for graphics and numeric work. B, C, and D are not processor workloads.
- Q23. A.** Pentium is an Intel family. B, C, and D are non-computer firms.
- Q24. A, B, C.** Intel, AMD, and ARM-architecture designers all make processor technology. Ink cartridges (D) are printer consumables.
- Q25. A, B, C.** The single-chip CPU, the GPU's parallel strength, and hertz clock units are correct. D is false: RAM is separate main memory.
- Q26. A.** A CPU is a general-purpose processor. GPUs specialize in parallel graphics/data workloads; a NIC and sound card are peripheral controllers.
- Q27. A.** A GPU has many parallel processing elements suited to graphics operations. Storage, input control, and firmware memory do not perform this computation.
- Q28. B.** A microcontroller integrates processing, memory, and peripherals for embedded control. The other choices do not describe that integrated design.
- Q29. A.** A washing-machine controller performs a dedicated embedded task. The other workloads typically use general-purpose or high-performance computer systems.
- Q30. A, B, C.** The first three describe processor roles. A GPU processes data; a hard disk stores it.